Human or Machine? Contrastive Learning for Detecting AI-Generated Chinese E-Commerce Reviews with a Custom Dataset

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Abstract

AI-generated content on Chinese e-commerce platforms undermines consumer trust and market fairness. Existing detectors trained on English corpora rarely generalize to the dense, slang-rich syntax of Chinese reviews or to cross-domain shifts. We introduce the first high-fidelity benchmark for this setting, pairing 13K human reviews with controlled synthetic counterparts produced through dual prompt strategies that preserve aspect-sentiment structure while diversifying surface style. The benchmark spans progressive out-of-distribution splits that vary generation models, paraphrasing style, and product domains, enabling granular robustness analysis. On top of strong supervised baselines and a representative zero-shot detector, we propose a hierarchical contrastive learning framework tailored to Chinese morpho-syntactic cues. The method leverages multi-level similarity objectives on bert-base-chinese encodings to capture organization- and model-specific signatures while remaining sensitive to human writing. It reaches 92.35 F1 in-domain and 50.72 under the most extreme shift, outperforming fine-tuned LLMs and encoder baselines. This study delivers actionable infrastructure and insights for trustworthy AI-generated content detection in high-stakes commercial ecosystems.

Keywords: AI-generated text detection, Chinese e-commerce, Cross-domain robustness, Contrastive learning, Large language models

1. Introduction

In 1950, Alan Turing proposed a test to evaluate machine intelligence: if a human evaluator could not distinguish between responses from a machine and a human, the machine could be considered intelligent (Turing, 1950). Today, as large language models (LLMs) achieve unprecedented fluency, researchers face the inverse challenge: distinguishing AI-generated content from human writing has become increasingly difficult, raising critical concerns about the integrity of digital information.

Recent leaps in large language model capability–from GPT-3's few-shot learning (Brown et al., 2020) to GPT-4's reasoning abilities (OpenAI, 2023)–have turned content generation into a high-fidelity mimicry task. Detection systems tuned on narrow distributions now break when confronted with novel domains, models, or writing styles, as comprehensive evaluations show (Zellers et al., 2019; Wu et al., 2023; Ji et al., 2025; Doughman et al., 2024).

Chinese e-commerce platforms amplify this risk: a $2.2 trillion market in 2023 with 11.6% projected CAGR where 93% of consumers consult reviews (Sharma, 2023; PowerReviews Team, 2023). Authenticity therefore determines both consumer trust and competitive fairness, yet automated generators offer inexpensive tools for flooding marketplaces with persuasive counterfeits.

Chinese e-commerce reviews mix traditional and simplified characters, emojis, transliterated brand names, and compact aspect-rich phrasing. The dense format and cultural terminology stress detectors trained on English news or academic corpora, while resources such as ASAP (Bu et al., 2021) lack paired synthetic counterparts required for authorship analysis.

Existing benchmarks like MGTBench and M4GT-Bench (He et al., 2024; Wang et al., 2024) focus on English narratives and do not probe progressive distribution shifts in Chinese commercial settings. As a result, we lack infrastructure for diagnosing robustness failures or designing Chinese-specific detection architectures.

This work addresses these gaps through three contributions:

- **Controlled Chinese benchmark.** We build a paired human–AI dataset via a controlled synthesis pipeline and benchmark mainstream Chinese LLMs for human-like writing to guide generator selection.

- **Robustness evaluation framework.** We derive three increasingly difficult out-of-distribution (OOD) test sets from the dataset and benchmark supervised fine-tuning plus zero-shot detectors to expose cross-domain resilience under extreme shifts.

- **Hierarchical contrastive learning.** We introduce a Chinese-optimized contrastive framework that sustains 92.35 and 50.72 F1 in-domain and under OOD-3, providing a deployable baseline.

Collectively, these contributions supply the infrastructure and models required to study trustworthy AI-generated content detection in Chinese e-commerce ecosystems.

2. Related Work

Efforts to build datasets for AI-generated text detection include broader benchmarks like TuringBench (Uchendu et al., 2021) and DeepfakeTextDetect (Crothers et al., 2022). More recent corpora such as HC3 (Guo et al., 2023) introduce paired human–AI responses, improving semantic parity, yet they often rely on a single model or limited prompt design, reducing stylistic diversity. These limitations mean detectors frequently learn superficial artifacts rather than robust authorship cues, motivating controlled synthesis approaches that enforce semantic alignment and stylistic variability as pursued in this study.

AI-generated text detection has progressed from statistical heuristics to fine-tuned Transformer models, with surveys documenting the landscape (Yang et al., 2025). Contemporary detectors typically adapt BERT or RoBERTa variants with auxiliary linguistic features (Devlin et al., 2019; Liu et al., 2019), yet they overfit source distributions and deteriorate when domains or generation models shift (Wang et al., 2024; Huang et al., 2024).

Analyses of syntactic templates and multilingual scenarios reinforce the brittleness of surface-pattern cues (Shaib et al., 2024). Contrastive learning addresses this by emphasizing relational signals (Jaiswal et al., 2021); DeTeCtive exemplifies hierarchical style modeling that motivates our design (Guo et al., 2024). Zero-shot detectors such as DetectGPT, Fast-DetectGPT, and Binoculars promise training-free deployment but remain validated mainly on English corpora (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024).

Chinese-focused research is sparse despite progress in foundational models like ERNIE and Chinese BERT (Sun et al., 2019; Cui et al., 2021). Commercial reviews add dense morphology, emojis, and domain terminology tied to recommendation systems (Ni et al., 2019), while datasets such as HC3 or ASAP emphasize sentiment over authorship (Guo et al., 2023; Bu et al., 2021). Consequently, prior work lacks a benchmark that captures Chinese e-commerce linguistics alongside cross-domain robustness diagnostics–gaps addressed in this study.

3. Data

The construction of a reliable benchmark dataset for Chinese AI-generated text detection requires systematic methodology that ensures high-fidelity AI-generated content while enabling comprehensive evaluation across different scenarios.

3.1. Design Principles

Our dataset design follows three linked goals: probe human-like abilities that matter in Chinese e-commerce reviews, stress-test generalization, and avoid contamination that could inflate scores. Existing benchmarks often miss these targets because direct, generic prompting introduces a "fidelity gap"—the synthetic reviews lack aspect-driven nuance and pragmatic intent, leaving detectors reliant on superficial artifacts that fail to transfer (Uchendu et al., 2021; Crothers et al., 2022). To close this gap, we adopt a controlled synthesis pipeline that keeps AI outputs thematically and sentimentally aligned with their human counterparts while injecting stylistic variation, pushing detectors to discriminate by style rather than topic (Guo et al., 2023).

3.2. Generation Pipeline

Prompt Generation Algorithm for Controlled Synthesis

procedure GeneratePrompts(D_{human})
  Input: D_{human}, dataset of human reviews with aspect sentiments
for each review R_i in D_{human} **do**
  Step 1 - Create Profile:
  $P \leftarrow$ Extract aspect sentiments from R_i
  $key \leftarrow$ Sort and join elements of P
  $R_i.profile \leftarrow$ Simplify(key)
  Step 2 - Generate Prompt:
  $details \leftarrow$ Parse($R_i.profile$)
  $T_{aspect}, T_{sent}, T_{other} \leftarrow$ Generate templates using $details, R_i.star$
  if $T_{aspect} \neq \emptyset$ and random() < 0.7 **then**
    $prompt \leftarrow$ RandomChoice(T_{aspect})
  else
    $prompt \leftarrow$ RandomChoice($T_{aspect} \cup T_{sent} \cup T_{other}$)
  end if
  $R_i.prompt \leftarrow prompt$
end for
return StratifiedSample(D_{human}, size=1000)

Table 1: Core prompt generation algorithm for controlled synthesis with dual-strategy template selection

The dataset construction process begins with the ASAP corpus (Bu et al., 2021), a large-scale collection of Chinese product reviews annotated with fine-grained aspect and sentiment labels. For each human-written review, the corresponding metadata, including overall star rating and aspect-level sentiment polarities (e.g., "Taste#Flavor," "Service#Queueing"), are extracted to form a comprehensive "emotional-semantic profile." Our controlled synthesis approach employs two complementary prompt generation strategies to maximize

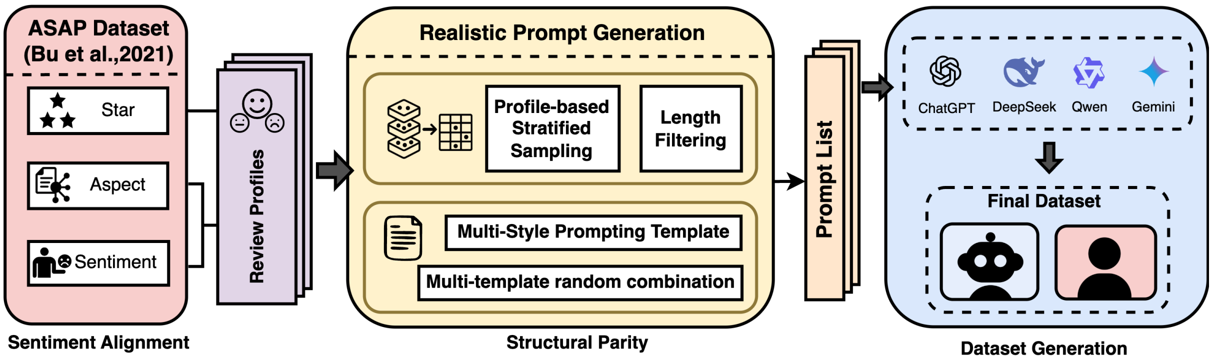

Figure 1: An overview of the dataset generation pipeline showing the systematic approach from human reviews to controlled AI synthesis

stylistic diversity while maintaining semantic coherence, ensuring comprehensive coverage of authentic review patterns.

The structured profiles are transformed into diverse, natural-language prompts using two complementary strategies. The first strategy employs profile-based templates that directly translate aspect-sentiment pairs into specific generation instructions, while the second uses style-transfer templates that ask models to imitate the writing patterns of existing human reviews. This dual-strategy framework ensures both semantic alignment and stylistic diversity in the generated content.

To ensure the fidelity and diversity of the AI-generated corpus while minimizing model-specific artifacts, we conducted the first systematic evaluation of mainstream LLMs for Chinese e-commerce review generation (detailed in Section 5.3). Based on comprehensive assessment across multiple dimensions, we selected optimal models for dataset construction using standardized prompt sets.

3.3. OOD Evaluation Framework

The resulting splits combine the multi-model training mix, an in-domain anchor, and three stepped OOD evaluations that mimic deployment extremes. The in-domain split keeps Deepseek-Chat-v3 food-review outputs in their original surface form, providing a clean baseline before stress testing, while scenario-specific prompt templates ensure each OOD suite reflects the intended shift.

- **In-domain baseline**: Same generator (Deepseek-Chat-v3), domain, and review style as training, yielding a direct read on overfitting.
- **OOD-1 model shift**: Swaps in GPT-4o-mini, Gemini-2.5-Flash, Deepseek-Chat-v3, and Qwen-Turbo while holding domain and style fixed, isolating generator variability.

Category	Templates
Direct Requests	写一个餐厅好评/差评 *Write a positive/negative restaurant review* 帮我写个餐厅评价 *Help us write a restaurant evaluation* 给餐厅写个评论/点评 *Write a restaurant comment/review*
Sentiment-Guided	写个正面的餐厅评价 *Write a positive restaurant evaluation* 写一个满意的餐厅评论 *Write a satisfied restaurant review* 给这家餐厅写个好评 *Write a favorable review for this restaurant*
Aspect-Specific	写个餐厅评价，说味道不错 *Write a review mentioning good taste* 评论服务态度很好的餐厅 *Review restaurant with excellent service* 点评价格合理的餐厅 *Review reasonably priced restaurant* 评价环境满意的餐厅 *Review restaurant with satisfactory environment*

Table 2: Profile-based prompt templates for controlled review generation across multiple semantic dimensions

- **OOD-2 style shift**: Adds paraphrasing from the same models to push machine outputs toward human phrasing, increasing near-source confusion.
- **OOD-3 domain shift**: Layers a 20% human-content blend onto OOD-2 and expands into electronics, clothing, and home-product reviews via the Amazon multilingual corpus (Keung et al., 2020), forcing cross-domain transfer.

Strategy	Templates
Style Mimicry	请基于以下真实评论重写一个类似的餐厅评价：{review}。要求：保持相同情感倾向，使用不同表达方式。 *Rewrite a similar restaurant review based on: {review}. Maintain emotional tone with different expressions.*
Content Variation	参考这个餐厅评论的风格，写一个内容不同的评价：{review}。注意：保持评价角度和语气一致。 *Reference this review's style for different content: {review}. Maintain consistent perspective and tone.*
Stylistic Imitation	模仿以下评论的写作风格，创作新的餐厅评价：{review}。要求：语调和表达习惯相似。 *Imitate the writing style to create new review: {review}. Maintain similar tone and expression patterns.*

Table 3: Style-transfer prompt templates for sophisticated linguistic pattern replication

Prompt Characteristic	Value	Coverage
Total Generated Prompts	10,000	100%
Unique Prompt Variants	8,333	83.3%
Profile-based Templates	5,000	50.0%
Rewrite-based Templates	5,000	50.0%
Distinct Template Patterns	20+	High diversity

Table 4: Comprehensive prompt generation statistics demonstrating balanced strategy distribution and high template diversity

Grounding every synthetic sample in a human counterpart and staging these shifts keeps aspect-controlled semantics intact while foregrounding stylistic cues over topical leakage, providing the backbone for the contrastive approach in Section 4.

3.4. Dataset Statistics and Analysis

Dataset	Human	Machine	Total
Training	8,000	8,000	16,000
Validation	1,000	1,000	2,000
Testing	1,000	1,000	2,000
Testing (OOD-1)	1,000	1,000	2,000
Testing (OOD-2)	1,000	1,000	2,000
Testing (OOD-3)	1,000	1,000	2,000
Overall	**13,000**	**13,000**	**26,000**

Table 5: Comprehensive dataset statistics across all evaluation scenarios

Each split remains class-balanced, giving detectors equal exposure to human and synthetic reviews while scaling from 16K training pairs to 2K samples in every evaluation condition. The staged OOD suites tie back to the controlled synthesis labels (aspect, sentiment, style) so that downstream objectives emphasize stylistic separability instead of topical overlap.

4. Methodology

Building on the comprehensive dataset and progressive evaluation framework established in Chapter 3, this chapter presents our methodological approach to Chinese AI-generated text detection. The hierarchical labeling system and controlled synthesis methodology introduced in our dataset construction directly inform the design of our detection framework, which leverages multi-level contrastive learning to capture the stylistic relationships identified in our data generation process.

4.1. Problem Formulation

We formalize AI-generated text detection as a binary classification task where the input consists of Chinese e-commerce reviews $\mathcal{X} = \{x_1, x_2, \ldots, x_n\}$ and corresponding labels $\mathcal{Y} = \{0, 1\}^n$ indicating human-written (0) or AI-generated (1) content. The optimization objective seeks to learn a robust classifier $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the expected risk under distribution shift, formally defined as $\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}_{target}}[\mathcal{L}(f_\theta(x), y)]$ where $\mathcal{D}_{target}$ represents the target domain distribution that may differ from the training distribution $\mathcal{D}_{source}$.

A key challenge is that many samples exhibit ambiguous characteristics near classification boundaries, where high-quality AI-generated content may closely resemble human writing styles. Traditional hard classification approaches may struggle with such boundary cases, motivating the exploration of soft classification methods that quantify uncertainty and provide nuanced decision-making capabilities.

4.2. Model Architecture Overview

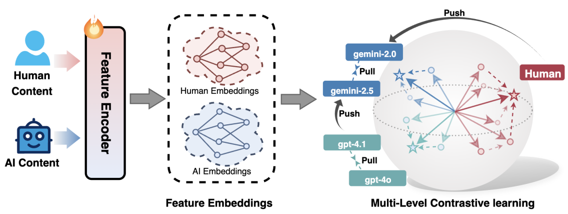

Figure 2: Overview of the improved multi-level contrastive learning model architecture showing token-level, span-level, and semantic-level representations

Our detection framework integrates supervised baselines with an enhanced multi-level contrastive encoder. Backbone comparisons rely on fine-tuning bert-base-chinese to capture the morpho-syntactic traits of Chinese reviews, while the proposed approach retains this encoder and layers on hierarchical contrastive losses to sharpen classification.

4.3. Baseline Contrastive Learning

The foundational contrastive learning approach employs the InfoNCE loss to learn discriminative representations by contrasting positive and negative sample pairs. The objective function is formulated as:

$$\mathcal{L}_{InfoNCE} = -\log \frac{\exp(\text{sim}(h_i, h_j^+)/\tau)}{\sum_{k=1}^{N} \exp(\text{sim}(h_i, h_k)/\tau)} \quad (1)$$

where h_i represents the anchor sample embedding, h_j^+ denotes the positive sample embedding, τ is the temperature parameter, and sim$(,)$ measures cosine similarity between representations. This baseline establishes the foundation for our enhanced multi-level approach.

4.4. Multi-Level Contrastive Learning

Core Intuition. Writing styles form hierarchical clusters (Guo et al., 2024); therefore we model each generator, organization, and the human corpus as progressively distant levels. Coupling controlled synthesis with a bert-base-chinese encoder captures the brevity, mixed scripts, and domain terminology characteristic of Chinese e-commerce reviews while remaining sensitive to human prose.

Optimization Perspective. Chinese LLMs share recognizable organizational signatures (Lyu et al., 2025; Kar et al., 2025), so the objective encourages high similarity within a model family, moderate similarity within an organization, and low similarity across organizations or between AI and human text. Treating author cohorts rather than individual samples stabilizes the contrastive signal and grounds the subsequent hierarchy design.

We aim to capture these kinship relations with a text encoder, allowing the encoder to capture the multi-level similarities and distinctions.

Hierarchy Design. Our approach defines four distinct levels of text distributions, ordered by their expected similarity: (1) P_1 - texts from the same specific LLM model (highest similarity), (2) P_2 - texts from LLMs developed by the same organization (high similarity), (3) P_3 - texts from any Chinese LLM (moderate similarity), and (4) P_4 - human-written texts (distinct from all AI texts). The key insight is that the model should learn to assign higher similarity scores to text pairs that are closer in this hierarchy.

Following established hierarchical similarity principles (Guo et al., 2024), we formulate the expected relationships for Chinese text distributions. Consequently, we expect the encoded features from different sources to reflect their hierarchical relations within the high-dimensional feature space as follows:

$$\mathbb{E}_{x\sim P_i, y\sim P_j}[Sim(\Phi(x), \Phi(y))] > \mathbb{E}_{x\sim P_i, y\sim P_{j+1}}[Sim(\Phi(x), \Phi(y))], \quad \forall 1 \le i \le j < 4, \quad (2)$$

where Sim denotes cosine similarity and $\Phi(\cdot)$ the encoder. The hierarchy is tailored to Chinese LLM ecosystems so that distributional proximity translates into higher similarity, encouraging the model to capture fine-grained relations across generator families and human prose.

Multi-level contrastive learning implementation. To implement this hierarchical similarity structure, we design a specialized labeling system and contrastive loss formulation.

Labeling Strategy. We adapt the hierarchical labeling strategy with minor modifications for Chinese model ecosystems, incorporating model-organization mappings specific to the Chinese LLM development landscape. Each text sample receives multiple labels: a primary label x_i indicating human (1) or AI-generated (0), an organization label y_i, and a model-specific label z_i. This enables our bert-base-chinese integration to effectively distinguish stylistic similarities across different hierarchy levels.

Mathematical Formulation. When processing a data batch containing N samples, for the ith sample T_i, we assign it with a label x_i. If the text is generated by an LLM, then $x_i = 0$, otherwise, $x_i = 1$. For those AI-generated text (i.e., $x_i = 0$), we further label the model series and the specific model with y_i and z_i. Then, the encoding function $\Phi(\cdot)$ maps the text into a d-dimensional feature space $\mathbb{R}^d$. For any two samples T_i and T_j, we compute the cosine similarity between their encoded features through $Sim(\Phi(T_i), \Phi(T_j))$, and define this similarity metric as $S(i, j)$. For human-written text $T_i(x_i = 1)$, the similarity of its encoding with other human-written text encodings should be greater than the similarity with AI-generated ones, hence the following relationship should be satisfied:

$$S(i, j) > S(i, k), \quad \forall x_j = 1, x_k = 0. \quad (3)$$

Similarly, for text $T_i(x_i = 0)$ generated by LLMs, Ineq. 1 suggests the existence of multi-level similarities and differences internally within LLMs, expressed as follows:

$$\begin{aligned} S(i,j) > S(i,l) > S(i,m) > S(i,n), \\ \forall z_i = z_j,\ z_i \ne z_l,\ y_i = y_l, \\ y_i \ne y_m,\ x_i = x_m,\ x_i \ne x_n. \end{aligned} \quad (4)$$

In order to achieve the above optimization objectives, we propose a method to solve these constraints hierarchically. Specifically, for the first inequality in Ineq. 3, we consider the index l, m, n that satisfies the conditions in the above constraints as a whole set, denoted as k, that is:

$$S(i,j) > S(i,k), \quad \forall z_i = z_j, z_i \neq z_k. \tag{5}$$

For the remaining inequalities, similar conditions are set to satisfy the constraints, culminating in a system that enforces hierarchical similarity:

$$\begin{cases} S(i,j) > S(i,k), & \forall x_i = 1, x_i = x_j, x_i \neq x_k \text{ (human texts more similar to each other)} \\ S(i,j) > S(i,k), & \forall x_i = 0, z_i = z_j, z_i \neq z_k \text{ (same model > different models)} \\ S(i,j) > S(i,k), & \forall x_i = 0, z_i \neq z_j, y_i = y_j, y_i \neq y_k \text{ (same org > different orgs)} \\ S(i,j) > S(i,k), & \forall x_i = 0, y_i \neq y_j, x_i = x_j, x_i \neq x_k \text{ (any AI > human).} \end{cases} \tag{6}$$

Contrastive Loss Design. To implement the hierarchical similarity constraints, we adapt the SimCLR framework with a key innovation: instead of single positive samples, we use *collections* of positive samples that meet specific hierarchical criteria. For any given sample, the positive set contains all samples at the same hierarchy level (e.g., all texts from the same model), while negative samples come from different hierarchy levels. This design encourages the model to learn that samples with closer hierarchical relationships should have higher similarity scores.

The contrastive learning loss averages similarity across all positive samples in the collection, ensuring stable training even with varying positive set sizes. The loss is formulated as:

$$\mathcal{L}_q = -\log \frac{\exp\left(\sum_{k\in K^+} \frac{S(q,k)}{\tau} / N_{K^+}\right)}{\exp\left(\sum_{k\in K^+} \frac{S(q,k)}{\tau} / N_{K^+}\right) + \sum_{k\in K^-} \exp\left(\frac{S(q,k)}{\tau}\right)}. \tag{7}$$

Multi-Level Loss Combination. The four hierarchical constraints result in four distinct contrastive losses ($\mathcal{L}_{q_i,1}, \mathcal{L}_{q_i,2}, \mathcal{L}_{q_i,3}, \mathcal{L}_{q_i,4}$), each operating at a different granularity level. The final loss combines these components with carefully balanced weights: for human texts, we apply only the human-vs-AI loss, while for AI texts, we apply all three AI-internal hierarchy losses. This design ensures that human samples are clearly distinguished from AI samples, while AI samples learn fine-grained distinctions within the AI hierarchy.

$$\mathcal{L}_{mcl} = \sum_{i=1}^{N} x_i \cdot (\delta \cdot \mathcal{L}_{q_i,1}) + (1 - x_i) \cdot (\alpha \cdot \mathcal{L}_{q_i,2} + \beta \cdot \mathcal{L}_{q_i,3} + \gamma \cdot \mathcal{L}_{q_i,4}). \tag{8}$$

Through this carefully designed multi-level framework, the model learns fine-grained stylistic features from different sources, enhancing both accuracy and generalization capabilities.

Multi-task auxiliary learning. Given that multi-task learning enables the model to simultaneously learn multiple tasks online by sharing useful information between different tasks, it promotes the model to learn more generic and discriminative features, hence enhancing the model's generalization ability. Therefore, based on the aforementioned contrastive learning framework, we integrate an MLP classifier into the output layer of the encoder. This classifier performs a binary classification to determine whether a given query text was generated by human or LLM. We introduce a cross-entropy loss $\mathcal{L}_{ce}$ to optimize this classifier as follows:

$$\mathcal{L}_{ce} = -\frac{1}{N} \sum_{i=1}^{N} x_i \cdot \log(p_i) + (1 - x_i) \cdot \log(1 - p_i), \tag{9}$$

where p_i is the probability of the ith sample x_i being classified as human-written. Therefore, the overall multi-task auxiliary multi-level contrastive loss is defined as:

$$\mathcal{L}_{all} = \mathcal{L}_{mcl} + \mathcal{L}_{ce}. \tag{10}$$

5. Experiments

5.1. Experimental Design

We benchmark a supervised decoder baseline (fine-tuned Llama-3-8B) together with three encoders—bert-base-chinese, chinese-roberta-wwm-ext, and a single-level contrastive bert-base-chinese—against the proposed multi-level contrastive detector across the progressive OOD splits, reporting accuracy, precision, recall, F1, and AUC averaged over three runs. To contrast supervised training with detection-by-prompt methods, we also evaluate the zero-shot Binoculars detector as a training-free baseline.

All models share the same training schedule (learning rate 2e-5, batch size 4, 10 epochs); contrastive variants add DeTeCtive-style objectives on top of the bert-base-chinese backbone for a controlled comparison.

5.2. Zero-shot Baseline Evaluation

We benchmark the zero-shot Binoculars detector (Hans et al., 2024) with the official implementation, sweeping validation thresholds before reporting the four evaluation scenarios summarized in Table 7.

The detector remains precision-oriented but recall collapses under distribution shifts, pushing F1 below 1% in OOD-2 and OOD-3. English-tuned zero-shot architectures therefore miss the stylistic cues that characterize Chinese e-commerce reviews.

5.3. LLM Generation Evaluation

The systematic evaluation of mainstream LLMs for Chinese text generation capability requires comprehensive assessment across multiple dimensions to identify optimal models for dataset construction.

Test Scenario	Metric	Llama-3-8B	BERT-Chinese	Chinese-RoBERTa	Contrastive Learning	Multi-Level Contrastive
In-Domain Test	Accuracy	80.65	88.15	85.40	86.95	**92.45**
	Precision	85.92	89.69	87.72	89.15	**93.18**
	Recall	80.40	88.15	85.40	86.95	**92.45**
	F1-Score	79.90	88.03	85.17	86.76	**92.35**
	AUC-ROC	0.926	0.961	0.954	0.943	**0.988**
OOD-1	Accuracy	79.40	83.90	80.05	80.10	**87.50**
	Precision	83.03	87.08	84.07	85.21	**89.98**
	Recall	80.40	83.90	80.05	80.10	**87.50**
	F1-Score	78.41	83.52	79.41	79.31	**87.29**
	AUC-ROC	0.919	0.934	0.93	0.911	**0.945**
OOD-2	Accuracy	58.85	68.95	63.85	64.80	**69.80**
	Precision	70.85	78.81	74.95	77.53	**81.17**
	Recall	58.85	68.95	63.85	64.80	**69.80**
	F1-Score	50.46	66.04	59.33	60.20	**66.77**
	AUC-ROC	0.819	**0.860**	0.850	0.805	0.743
OOD-3	Accuracy	53.45	54.30	50.10	50.80	**58.80**
	Precision	68.24	74.50	75.03	75.20	**75.59**
	Recall	53.45	54.30	50.10	50.80	**58.80**
	F1-Score	40.57	42.43	33.56	35.09	**50.72**
	AUC-ROC	0.739	0.577	0.713	0.748	**0.807**

Table 6: Comprehensive performance comparison across all evaluation metrics and test scenarios. Shows results for six models across four test datasets with five evaluation metrics. **Bold** indicates best performance and underline indicates second-best performance for each metric and test scenario.

Test Set	Acc	Prec	Rec	F1	AUC
In-domain	65.75	99.68	31.60	47.99	6.98
OOD-1	54.80	98.89	8.97	16.45	14.44
OOD-2	50.10	75.00	0.30	0.60	22.75
OOD-3	49.90	40.00	0.40	0.79	38.15

Table 7: Zero-shot Binoculars performance (no supervised training on our dataset). Metrics are reported as percentages after validation-driven threshold optimization.

Table 8 ranks generation models; DeepSeek-Chat-v3 best balances lexical diversity, semantic cohesion, and distributional similarity, so it anchors the synthetic corpus.

Based on the evaluation results, we construct the dataset with higher representation from better-performing models while maintaining sufficient diversity across different generation approaches.

5.4. Detection Performance Analysis

Table 6 compiles the supervised detectors, while Table 7 situates the zero-shot baseline for contrast. The sharp drops in OOD-2 and OOD-3 are anticipated: these splits deliberately stack paraphrasing and cross-domain shifts to create stress-test environments that reward genuine stylistic robustness rather than topical overlap, so lower absolute scores signal the intended difficulty.

Parameter scale helps in benign settings—fine-tuning Llama-3-8B reaches 79.90 F1 in-domain—but the proposed multi-level contrastive detector, built on the lighter bert-base-chinese encoder, delivers 92.35 F1 in-domain and still holds 50.72 F1 under OOD-3, surpassing Llama-3-8B (40.57) and single-level baselines (42.43/33.56). Hierarchical contrastive supervision therefore recovers large-model robustness without incurring large-model costs. Paired t-tests across three runs confirm the gains in every OOD split ($p < 0.001$, Cohen's $d > 0.8$) (Cohen, 1988).

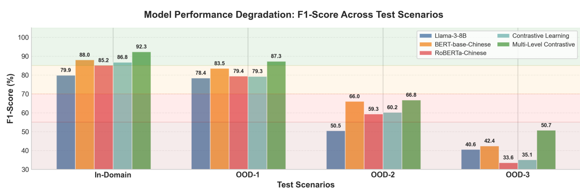

Figure 3: F1-score degradation across progressive out-of-distribution scenarios (In-Domain, OOD-1, OOD-2, OOD-3). Error bars represent standard deviation across 3 independent runs.

Figure 3 visualizes this trajectory, underscoring how the multi-level contrastive detector degrades more gracefully as the evaluation progresses from model shifts to paraphrasing and finally to the cross-domain stress test.

Model	TTR	Avg Length	Self-BLEU	Self-BERTScore	JSD vs Human	Rank
Human	0.0923	283.23	0.1214	0.0007	0.0000	–
deepseek-chat-v3-0324	**0.0376**	**202.64**	**0.5258**	**0.0074**	**0.4131**	**1**
qwen-turbo	0.0281	183.11	0.5932	0.0097	0.4188	2
gemini-2.0-flash-001	0.0285	102.40	0.5842	0.0100	0.4196	3
gpt-4o-mini	0.0216	157.39	0.6281	0.0146	0.4527	4

Table 8: Comprehensive benchmark evaluation demonstrating DeepSeek-Chat-v3's superiority in generating human-like Chinese e-commerce reviews

5.5. Ablation Study

To systematically understand the contribution of different components in our multi-level contrastive learning framework, we conduct ablation studies examining the progressive integration of key architectural components. We report results on the OOD-1 dataset, which best demonstrates the performance differences between different model configurations under moderate distribution shifts.

Configurations	Accuracy	Precision	Recall	F1
$\mathcal{L}_{all}$ (Full Model)	**87.50**	**89.98**	**87.50**	**87.29**
$\mathcal{L}_{pcl} + \mathcal{L}_{ce}$	84.35	88.06	84.35	83.94
w/o $\mathcal{L}_{ce}$	85.95	89.01	85.95	85.65
w/ α = 0	86.30	89.14	86.30	86.03
w/ β = 0	86.15	88.14	86.15	85.86
w/ γ = 0	85.50	88.74	85.50	85.17
w/ classification head	84.80	84.65	84.75	84.65

Table 9: Ablation study demonstrating progressive performance improvement through multi-level contrastive learning components

The ablation indicates that removing cross-entropy or individual hierarchy weights reduces F1 by one to three points, whereas the full objective maintains 87.29 F1 on OOD-1, confirming the benefit of combining multi-level signals.

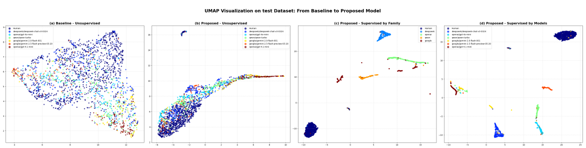

Figure 4: UMAP dimensionality reduction of the test dataset comparing baseline and proposed models under different supervision levels

Embedding Visualization. UMAP projections in Figure 4 show clearer separation under the proposed hierarchy than the baseline, reinforcing the learned stylistic structure.

6. Discussion

The stress test OOD suite shows that Chinese AI text detection departs from English centric assumptions: style transfer and cross domain shifts push every system well below in domain scores, and only the hierarchically supervised detector retains useful signal. This highlights the need to capture Chinese specific cues such as mixed scripts, aspect bundling, and pragmatic idioms, and to favour models that encode these features over topical overlap or large decoder capacity.

Interpretability still trails performance. Binoculars illustrates how opaque scoring struggles with Chinese typography, indicating that detectors should expose stylistic rationales that auditors can verify. Watermarking or provenance tagging offers a complementary route that can supply stronger evidence when detectors flag ambiguous reviews. Combining explainable contrastive models with provenance aware pipelines is therefore a promising next step toward trustworthy Chinese e commerce moderation.

7. Conclusion

This study delivers a controlled Chinese e-commerce benchmark that couples human reviews with stylistically aligned synthetic counterparts and stresses detectors through progressive OOD splits. Leveraging this resource, a hierarchical contrastive framework proves that targeted supervision can unlock robust, interpretable detection while situating encoder, decoder, and zero-shot baselines on common ground. Together, the dataset, methodology, and analysis offer a practical foundation for future work on trustworthy AI-generated content moderation in Chinese markets and related domains.

8. Ethics Statement and Limitations

The benchmark remains bounded by today's training data. Even with controlled synthesis, detectors inherit the coverage of the source reviews; once marketplace slang, generator families, or attack tactics evolve, performance will decay unless corpora are refreshed. Zero-shot detectors offer an appealing alternative yet react sharply to hidden model updates, language shifts, and unseen generators—especially in Chinese settings where stylistic cues are underexplored. Future work must pair continual dataset renewal with Chinese-oriented zero-shot research so that detection keeps pace without assuming prior access to every frontier model.

9. Data and Code Availability

The code and datasets used in this work will be released upon publication.

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